PUMPING WATER BY COMPRESSED AIR.

THOSE interested in mining and hydraulics will be glad to have some account of the Pohlé system of raising water from non-flowing wells, mines, etc., which has been introduced into Canada by the Ingersoll-Sergeant Drill Company, of Montreal.

The following description is given by the makers:

The pump proper consists of only two plain open-ended pipes, the larger one with an enlarged end piece constituting the discharge pipe, and the smaller one let into the enlarged end piece of the discharge pipe constitutes the air inlet pipe, through which the compressed air is conveyed to the enlarged end piece to the under side of the water to be raised. No valves, buckets, plungers, rods or other moving parts are used within the pipes or well.

In pumping, compressed air is forced through the air pipe into the enlarged end at the bottom of the water pipe, thence by the inherent expansive force of the compressed air, layers or pistons of air are formed in the water pipe, which lift and discharge the water layers through the upper end of the water discharge pipe. At the beginning of the operation the water surface outside of the pipe and the water surface inside of the pipe are at the same level; hence the vertical pressures per square inch are equal at the submerged end of the pipe, outside and inside. As air is forced into the lower end of the water pipe, it forms alternate layers with the water, so that the pressure per square inch of the column thus made up of air and water, as it rises inside of the water pipe, is less than the pressure of water per square inch outside of the pipe.

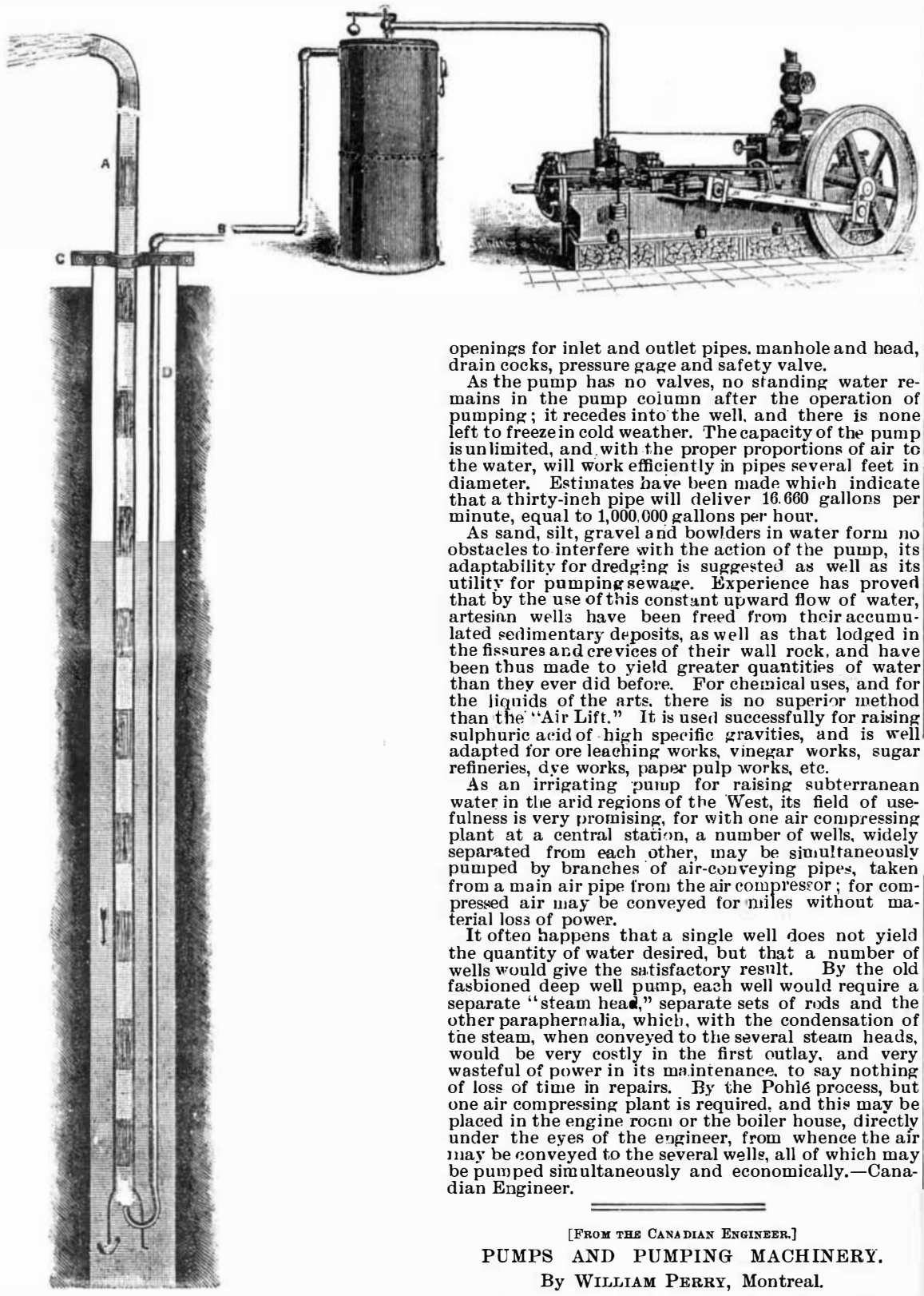
Owing to this difference of pressure, the water flows continually from the outside to within the water pipe by gravity force, and its ascent through the pipe is free from shock, jar or noise of any kind.

These air sections or strata of compressed air form watertight bodies, which, in their ascent in the act of pumping, permit no "slipping" or back flow of water. As each air stratum progresses upward to the spout, it expands on its way in proportion as the overlying weight of water is diminished by its discharge, so that the air section, which may have been say 50 pounds per square inch at first, will be only 1·74 pounds when it underlies a water layer of four feet in length at the spout, until finally this air section, when it lifts up and throws out this four feet of water, is of the same tension as the normal atmosphere; thus proving that the whole of its energy was used in work, and that this pump is a perfect expansion engine.

As the weight of the water outside of the discharge pipe (the head) is one-third greater per square inch than the aggregate water sections within the pipe when in operation, it follows that the energy due to this one-third greater weight is utilized in overcoming the resistance of entry into the pipe, and all the friction within it.

The Pohlé "air lift" pump gives ninety per cent. of efficiency from the air receiver in water pipes of large diameter, and as a rule, above eighty per cent. It retains this efficiency without repairs, or until the pipes rust through, whereas ordinary bucket and plunger pumps gradually lose efficiency from the first stroke they make, and lose it rapidly if the water contains sand or is acid in character. It has been estimated by competent experts, that under favorable conditions and large diameters of water and air pipes, 1,000,000 gallons of water can be raised 100 feet high with one and a half tons of good coal.

The air reservoirs are all strongly made of homogeneous steel, tested and guaranteed at working pressures of 110 pounds; they are provided with the proper openings for inlet and outlet pipes, manhole and head, drain cocks, pressure gage and safety valve.



As the pump has no valves, no standing water remains in the pump column after the operation of pumping; it recedes into the well, and there is none left to freeze in cold weather. The capacity of the pump is unlimited, and with the proper proportions of air to the water, will work efficiently in pipes several feet in diameter. Estimates have been made which indicate that a thirty-inch pipe will deliver 16,660 gallons per minute, equal to 1,000,000 gallons per hour.

As sand, silt, gravel and bowlders in water form no obstacles to interfere with the action of the pump, its adaptability for dredging is suggested as well as its utility for pumping sewage. Experience has proved that by the use of this constant upward flow of water, artesian wells have been freed from their accumulated sedimentary deposits, as well as that lodged in the fissures and crevices of their wall rock, and have been thus made to yield greater quantities of water than they ever did before. For chemical uses, and for the liquids of the arts, there is no superior method than the "Air Lift." It is used successfully for raising sulphuric acid of high specific gravities, and is well adapted for ore leaching works, vinegar works, sugar refineries, dye works, paper pulp works, etc.

As an irrigating pump for raising subterranean water in the arid regions of the West, its field of usefulness is very promising, for with one air compressing plant at a central station, a number of wells, widely separated from each other, may be simultaneously pumped by branches of air-conveying pipes, taken from a main air pipe from the air compressor; for compressed air may be conveyed for miles without material loss of power.

It often happens that a single well does not yield the quantity of water desired, but that a number of wells would give the satisfactory result. By the old fasbioned deep well pump, each well would require a separate "steam head," separate sets of rods and the other paraphernalia, which, with the condensation of the steam, when conveyed to the several steam heads, would be very costly in the first outlay, and very wasteful of power in its maintenance, to say nothing of loss of time in repairs. By the Pohlé process, but one air compressing plant is required, and this may be placed in the engine room or the boiler house, directly under the eyes of the engineer, from whence the air may be conveyed to the several wells, all of which may be pumped simultaneously and economically.—Canadian Engineer.

[FROM THE CANADIAN ENGINEER.]

PUMPS AND PUMPING MACHINERY.

By WILLIAM PERRY, Montreal.

THE date from which we commence the history of pumps is the year 200 B. C. Previous to that period there is no mention made of them. Nor has there been discovered any portion that can be judged appertaining to such a machine; the heathen Chinese cannot claim any priority in this special branch, which is peculiar, the more so when we consider their manner of irrigation. A pump is even now a rarity with them.

A pump is not a very intricate machine in itself, and its parts are comparatively few. But its action, or want of it, sometimes makes it seem most mysterious. To those readers who have not considered the question of water dispensation, the remark that pumping machinery stands prominent among the various branches of engineering may seem to allow of discussion. A few instances will very soon give ample proof. How could our coal be obtained and our mines worked, if not for the pumping plant? Our water supply obtained, or our sewage and chemical works carried on? When man enters Nature's storehouse in search of wealth, he finds water ever ready to dispute his supremacy; it may be in a constant stream, varying only with the season; oftentimes vast quantities are stored in crevices of the rocks. Some idea of the quantity of water raised will be given when it is known that often its weight is double and treble that of other material raised, and is frequently 30 per cent. in coal mines.

Rude nations have not possessed the machine, simple as it is, but have always resorted to a more laborious method to obtain water. In the early ages it does not appear to have been known to the Greeks or Romans. Although the pump was invented 200 B. C., it was not until the beginning of the seventeenth century that its true principles were understood; although in 1636 fire engines were built in Holland, and from which, as far as general principles are concerned, no improvements have been made. In 1582, Peter Morris, a Dutchman, contrived a water engine to supply the residents of London from the Thames, and threw water over the steeple of St. Magnus church, at the north end of London Bridge. The introduction of machinery for domestic use commences from May, 1582. The first patent record in England of a pump is patent No. 6, year 1618, by David Ramsay and Thomas Wildgoose. The air pump was invented by Otto de Guericke in 1654. In 1660 Robert Boyle made many improvements in the air pump.

Sir Samuel Morland, Master of Mechanics to King Charles II, in the year 1674 invented and patented the plunger pump, made of cast iron in 1675. He threw water 60 feet high at the rate of 60 barrels per hour, with 8 men, in 1681; the king presented him with a medallion portrait set in diamonds. 1695 is the first notice I have of ship pumps. In 1732 Mr. Demoun invented a pump like a V. In 1741 James Creed secured a patent, No. 579, for making three different machines for making lead pipe for pump use. The first oscillating pump was patented in the year 1750, patent No. 658, by W. Perkins.

The pumps commonly used for raising water from wells may be divided into two classes—lifting pumps and forcing pumps. The lifting pumps may be again subdivided into two varieties, namely, those with a hollow piston and those with a solid or plunger piston.

1. Lifting pumps with a hollow piston, called also atmospheric pumps. This variety, in its simplest form, consists of the following parts:

A cylinder or tube, in which is fixed a valve opening upward, and above which works a piston provided with a valve also opening upward. The part of the cylinder in which the piston works is called the body of the pump, and is the only part which need be bored with any great accuracy. The top of the cylinder may be opened or closed, it matters not which, but somewhere above the level to which the piston ascends there must be an orifice for discharging the water.

The action of the common atmospheric pump is so simple, and is so well known to every school boy, that it will be unnecessary here to dwell upon it. The cylinder is made of various materials, as wood, iron, or copper, and frequently the lower part below the fixed valve is a mere iron pipe furnished with a strainer at its lower extremity. The fixed valve in this kind of pump must be placed at such a level that the depth from it to the surface of the water in the well must never exceed the height of a column of water which will balance the atmospheric pressure or weight of the atmosphere. This weight is measured in the barometer by a column of mercury, which varies in different parts of the world, and at different altitudes, from 28 to 31 inches. Thus, an atmospheric pump at the level of the sea may have its fixed valve several feet higher than a similar pump working on the top of a high mountain. The height at which the mercury stands in a barometer at any given place affords, in fact, a tolerably practical measure of the height to which water will rise in a vacuum when pressed by the external atmosphere. Thus, in theory, where the mercury stands in the tube of a barometer at a height of 30 inches, the sucker or fixed valve of an atmospheric pump may be placed 30 feet above the surface of water in a well. In practice, however, owing to imperfection of materials, fluctuations of level in the water, and other causes, this difference of level is too great, and should not really exceed 25 feet. In shallow wells, therefore, which are not more than about 27 feet in depth, the part of the cylinder or pump above the fixed valve need never exceed the length of the slope or space through which the piston works. In deep wells the ascending part of the cylinder above the body of the pump in which the piston works may be, theoretically, of any height. There are difficulties, however, connected with the valves in the movable piston which render it inconvenient to have the lift in this kind of pump much more than 100 feet. Whatever may be the height of the column of water above the movable piston, it is evident that the absolute weight of this whole column has to be lifted at each stroke of the piston, and for this reason atmospheric pumps, which are worked by hand, have scarcely any of the pump above the piston, as otherwise the weight of water to be lifted at each stroke would be too great for the power to be applied. This practically limits the height to which water can be raised from wells by common atmospheric pumps worked by hand to about 25 feet.

In deep wells, however, when pumps are worked by horse or steam power this objection does not apply, and if the power be sufficient to raise at each stroke the whole column of water above the piston, the length of the cylinder above this piston is only limited by the practical considerations before alluded to in connection with the valves. It should be observed that the common atmospheric pump is seldom or never used in waterworks for the purpose of raising water.

ON CALCULATING THE POWER OF PUMPING ENGINES.

The work performed by steam engines is commonly expressed in what is termed "horse power," that is, an engine is said to be equal to the work performed by a certain number of horses. The standard which has been fixed on to represent the work of one horse is equal to 33,000 pounds raised through a space of one foot high in a minute. This is equivalent to saying that a horse walking at his most effective speed of 2½ miles an hour, or 220 feet per minute, and attached to a weight of 150 pounds freely suspended over a pulley, will raise this weight at the same rate of 220 feet per minute. Using, then, this standard for computing the work of engines—a standard which has been agreed to by the mechanicians of all countries—we obtain a very ready method of determining the horse power required to raise any given quantity of water to any required height. The data required for this purpose

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